MAT221 Midterm Examination, Fall 2024

KU Email: _____

Name (Korean or English): _____

Student ID # _____

Date	Thursday, October 24th
Time Period	1:30-2:45pm
Duration of Test	75 minutes
Number of Test Pages	8 pages (4 double-sided sheets) in total
Total Possible Marks	42
Additional Materials Allowed	None. NO CALCULATORS.

Instructions

1. Answer the questions in the spaces provided. If you require additional space to answer a question, please use the extra page(s) and refer to this page in your solutions.
2. Your grade will be influenced by how clearly you express your ideas, and how well you organize your solutions.
3. **This exam will be scanned and marked electronically.** Be sure to write legibly and **make sure that your writing isn't too faint / light.**
4. DO NOT WRITE NOTES/FORMULAS ON THIS COVER PAGE. If you do, you may get ZERO.
5. Writing any language other than **ENGLISH** on a problem will result **ZERO** on that problem. Writing other languages on a scratch work is okay, but you need to indicate which work is a scratch work and which is not.
6. Convention: $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, vector spaces are assumed to be $\mathbb{F}$-vector spaces, and column vectors can be rewritten as a tuple (not a row vector), i.e.,

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = (a_1, a_2, \ldots, a_n).$$

1. (10 points) Consider the following set

$$W := \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \ : \ f(x) = f(-x) \quad \forall x \in \mathbb{R}\} \subseteq \mathcal{F}(\mathbb{R}, \mathbb{R}),$$

which is the subset of even functions $\mathbb{R} \to \mathbb{R}$. Prove that W is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Consider zero-fcn $0 : \mathbb{R} \to \mathbb{R}$. Then $0(x) = 0 = 0(-x) \quad \forall x \in \mathbb{R} \Rightarrow \underline{0 \in W}$.

$\forall f, g \in W, \quad \forall x \in \mathbb{R}, \quad (f+g)(x) = f(x) + g(x) = f(-x) + g(-x) = (f+g)(-x)$

$\Rightarrow \underline{f+g \in W}$.

$\forall f \in W, \quad \forall c \in \mathbb{R}, \quad \forall x \in \mathbb{R}, \quad (cf)(x) = c \cdot f(x) = c \cdot f(-x) = (cf)(-x) \Rightarrow \underline{cf \in W}$.

By a Thm from the class, those 3 conditions $\Rightarrow$ W is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$. □

2. (10 points) Determine whether the following is a basis of $\mathbb{F}^3$ or not:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 5 \\ 8 \end{bmatrix}$$

Sol 1: $v_1 \neq 0$, $cv_1 = \begin{bmatrix} c \\ c \\ c \end{bmatrix} \neq \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = v_2 \quad \forall c \in \mathbb{F} \Rightarrow v_1, v_2$ lin indep.

If $v_3 \in \text{span}(v_1, v_2)$, then $\exists a, b \in \mathbb{F}$ s.t. $v_3 = av_1 + bv_2$

$$\Rightarrow \begin{bmatrix} a+b \\ a+2b \\ a+3b \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 8 \end{bmatrix} \begin{matrix} (1) \\ (2) \\ (3) \end{matrix} \overset{(1)}{\Rightarrow} b = 1-a \overset{(2)}{\Rightarrow} a + 2(1-a) = 5 \Rightarrow 2 - a = 5 \Rightarrow a = -3$$

$$\Rightarrow b = 1 - (-3) = 4.$$

Check (3) on $(a, b) = (-3, 4)$: $(-3) + 3(4) = 8$

$9 = 8$, ※

$\Rightarrow$ no such a, b exists $\Rightarrow v_3 \notin \text{span}(v_1, v_2) \Rightarrow \underbrace{v_1, v_2, v_3}$ lin indep.

$\dim \mathbb{F}^3 = 3$ vectors

$\Rightarrow v_1, v_2, v_3$ basis of $\mathbb{F}^3$. □

Sol 2: For lin indep of v_1, v_2, v_3, check $av_1 + bv_2 + cv_3 = 0$

$$\Rightarrow \begin{bmatrix} a+b+c \\ a+2b+5c \\ a+3b+8c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

$(1) \Rightarrow c = -a-b \overset{(2)}{\Rightarrow} a + 2b + 5(-a-b) = 0 \Rightarrow -4a - 3b = 0$

$$\Rightarrow b = -\frac{4}{3}a \Rightarrow c = -a - b = -\frac{7}{3}a$$

$$(3) \Rightarrow a + 3\left(-\frac{4}{3}a\right) + 8\left(-\frac{7}{3}a\right) = 0 \Rightarrow -\frac{65}{3}a = 0 \Rightarrow a = 0$$

$$\Rightarrow b = -\frac{4}{3} \cdot 0 = 0, \quad c = -a - b = 0$$

$\Rightarrow a = b = c = 0$ only soln $\Rightarrow \underbrace{v_1, v_2, v_3}$ lin indep

$\dim \mathbb{F}^3 = 3$ vectors

$\Rightarrow v_1, v_2, v_3$ basis of $\mathbb{F}^3$. □

3. (10 points) Consider the following linear transformation:

$$T\colon \mathbb{F}^4 \to \mathbb{F}^2$$
$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \mapsto \begin{bmatrix} x-w \\ 3x-2y+z+w \end{bmatrix}$$

Find a basis of $\ker(T)$.

$$\ker(T) = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{F}^4 : \begin{bmatrix} x-w \\ 3x-2y+z+w \end{bmatrix} = 0 \right\} = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{F}^4 : x=w,\ 4x-2y+z=0 \right\}$$

$$= \left\{ \begin{bmatrix} x \\ y \\ -4x+2y \\ x \end{bmatrix} \in \mathbb{F}^4 : x, y \in \mathbb{F} \right\} = \left\{ x \begin{bmatrix} 1 \\ 0 \\ -4 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} : x, y \in \mathbb{F} \right\}$$

$$= \operatorname{span}\left(\begin{bmatrix} 1 \\ 0 \\ -4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} \right).$$

$$\begin{bmatrix} 1 \\ 0 \\ -4 \\ 1 \end{bmatrix} \neq 0, \quad c \begin{bmatrix} 1 \\ 0 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} c \\ 0 \\ -4c \\ c \end{bmatrix} \neq \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} \quad \forall c \in \mathbb{F} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ -4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} \text{ lin indep}$$

$$\Rightarrow \boxed{\begin{bmatrix} 1 \\ 0 \\ -4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} \text{ basis of } \ker(T).}$$

4. Consider a basis $\beta = \{1, x-2, (x-2)^2\}$ of $\mathrm{P}_2(\mathbb{F})$.

(a) (6 points) Compute $[x^2+x+1]_\beta$, which is a coordinate vector in $\mathbb{F}^3$.

$x^2+x+1 = (x^2-4x+4)+5x-3 = (x-2)^2+5(x-2)+7\cdot 1$

$\Rightarrow [x^2+x+1]_\beta = \begin{bmatrix} 7 \\ 5 \\ 1 \end{bmatrix}$

(b) (6 points) Consider a linear transformation $T\colon \mathrm{P}_2(\mathbb{F}) \to \mathrm{P}_2(\mathbb{F})$ defined by $T(x^m) := (2-m)x^m$ for all $m = 0, 1, 2$. Compute the matrix representation $[T]_\beta$ of T.

$T(1) = T(x^0) = (2-0)x^0 = 2\cdot 1$

$\Rightarrow [T(1)]_\beta = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$

$T(x-2) = T(x^1) - 2T(1) = (2-1)x^1 - 2\cdot 2 = x-4 = (x-2)-2\cdot 1$

$\Rightarrow [T(x-2)]_\beta = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

$T((x-2)^2) = T(x^2-4x+4) = T(x^2) - 4T(x^1) + 4T(1)$

$= (2-2)x^2 - 4x + 4\cdot 2 = -4x+8 = -4(x-2)$

$\Rightarrow [T((x-2)^2)]_\beta = \begin{bmatrix} 0 \\ -4 \\ 0 \end{bmatrix}$

$\Rightarrow [T]_\beta = \begin{bmatrix} 2 & -2 & 0 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix}$

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Clearly state the question number being answered and refer the marker to this page.

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